

■ 1.2.2.1.1 Control and Display Panels. (See figure 1-7.)

The crew compartment has 12 control and display panels: two main display panels (1 and 2) that are canted forward 10° , two center panels (3 and 4) that slope down and aft 45° towards the horizontal, two bottom side panels (5 and 6), two lower side panels (8 and 12), one center side panel (14), two upper side panels (11 and 16), and the orbital rate display - earth and lunar (ORDEAL) panel aft to panel 8.

Panels 1 and 2 are located on each side of the front face assembly centerline, at eye level. Each panel is constructed of two 0.015-inch-thick aluminum-alloy face sheets, spaced 2 inches apart by formed channels. The spacer channels are located along the sheet edges; additional channels, inboard of the edge channels, reinforce the sheets. This forms a rigid box-like construction with a favorable strength-to-weight ratio and a relatively high natural frequency. Four shock mounts support each panel on the structure. Panel instruments are mounted to the back surface of the bottom and/or to the top sheet of the panel. The instruments protrude through the top sheet of the panel. All dial faces are nearly flush with the forward face of the panel. Panel 1 contains warning lights, flight indicators and controls, and propellant quantity indicators. Panel 2 contains caution lights, flight indicators and controls, and Reaction Control Subsystem (RCS) and Environmental Control Subsystem (ECS) indicators and controls.

Panel 3, immediately below panels 1 and 2, spans the width of these two panels. Panel 3 contains the radar antenna temperature indicators and engine, radar, spacecraft stability, event timer, RCS and lighting controls.

Panel 4 is centered between the flight stations and below panel 3. Panel 4 contains attitude controller assembly (ACA) and thrust translation controller assembly (TTCA) controls, inertial subsection indicators, and LM guidance computer (LGC) indicators and controls. Panels 1 through 4 are within easy reach and scan of both astronauts.

Panel 5 and 6 are in front of the flight stations at astronaut waist height. Panel 5 contains lighting and mission timer controls, engine start and stop pushbuttons, and the X-translation pushbutton. Panel 6 contains abort guidance controls.

■ Panel 8 is at the left of the Commander's station. The panel is canted up 15° from the horizontal; it contains controls and displays for explosive devices, audio controls, and heater controls.

Panel 11, directly above panel 8, has five angled surfaces that contain circuit breakers. Each row of circuit breakers is canted 15° to the line of sight so that the white band on the circuit breakers can be seen when they open.

Panel 12 is at the right of the LM Pilot's station. The panel is canted up 15° from the horizontal; it contains audio, communications, and communications antennas controls and displays.

Panel 14, directly above panel 12, is canted up 36.5° from the horizontal. It contains controls and displays for electrical power distribution and monitoring.

Panel 16, directly above panel 14, has four angled surfaces that contain circuit breakers. Each row of circuit breakers is canted 15° to the line of sight so that the white band on the circuit breakers can be seen when they open.

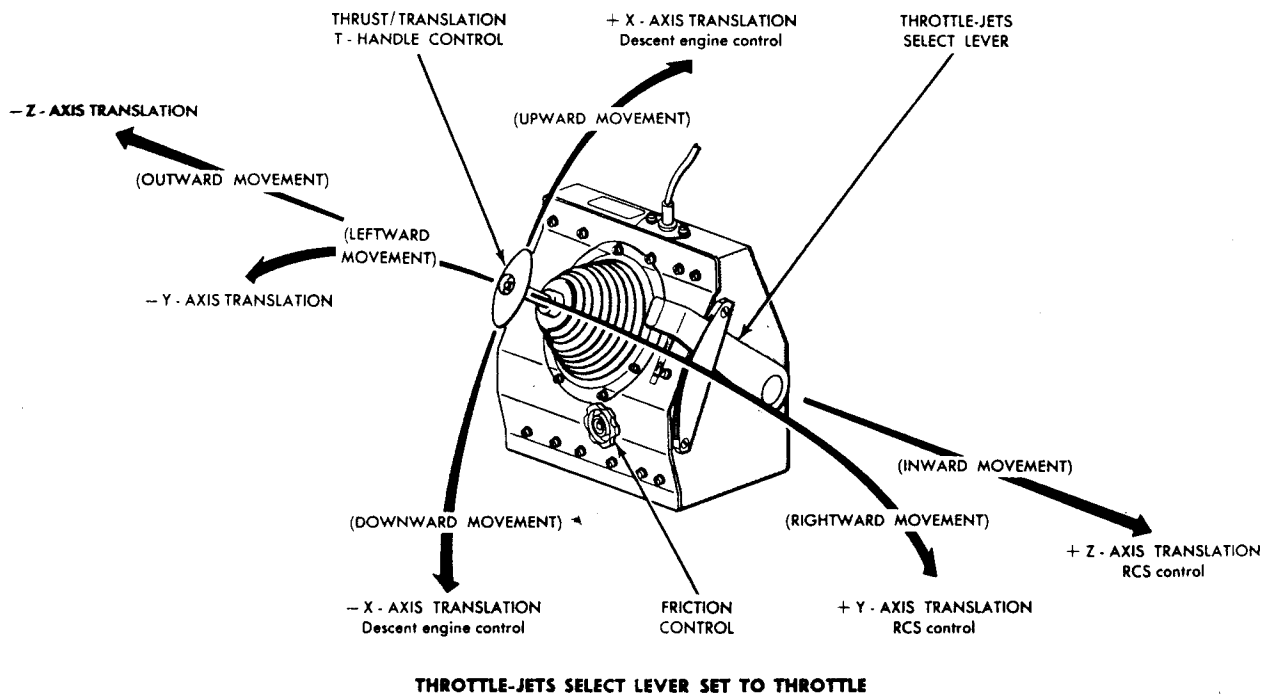
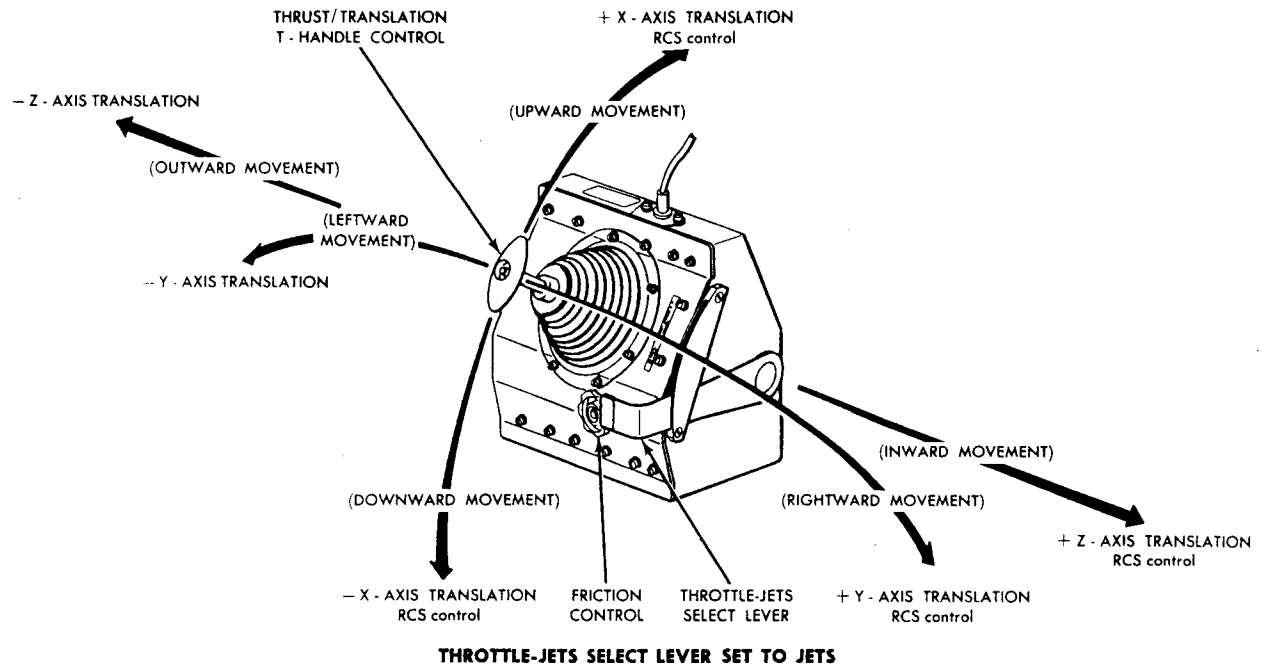
The ORDEAL panel is immediately aft of panel 8. It contains the controls for obtaining LM attitude, with respect to a local horizontal, from the LGC.

1.2.2.1.2 Forward Hatch. (See figure 1-8.)

The forward hatch is in the front-face assembly, just below the lower display panels. The hatch is approximately 32 inches square; it is hinged to swing inboard on quick-release hinge pins when opened. A cam latch assembly holds the hatch in the closed position; the assembly forces a lip, around the outer circumference of the hatch, into a preloaded elastomeric silicone compound seal that is secured to the LM structure. Cabin pressurization forces the hatch lip further into the seal, ensuring a pressure-tight contact. A handle is provided on both sides of the hatch, for latch operation. To open the hatch, the cabin must be completely depressurized by opening the cabin relief and dump valve on the hatch. When the cabin is completely depressurized, the hatch can be opened by rotating the latch handle. A lockpin in a plate over the latch can be withdrawn to release the latch in an emergency. The cabin relief and dump valve can also be operated from outside the LM.

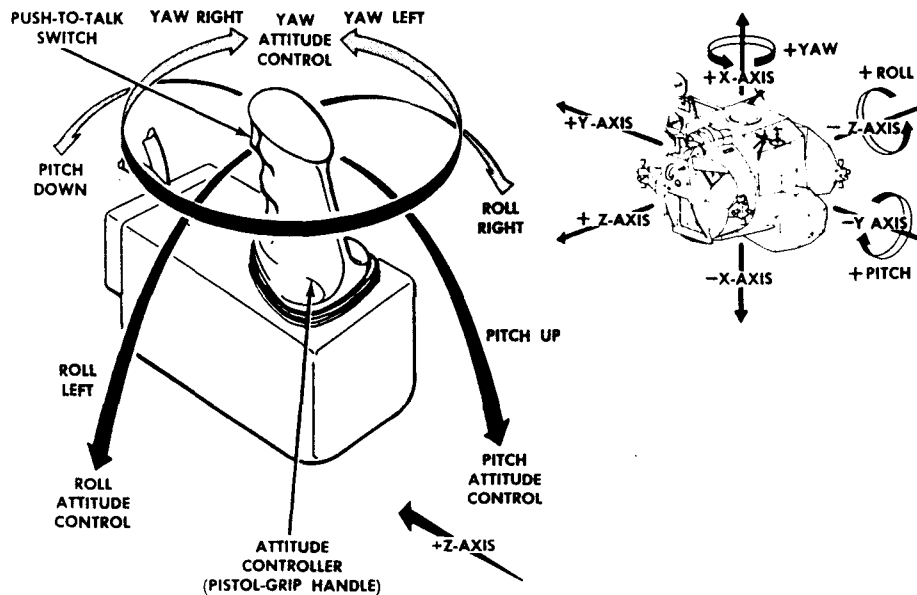
ASCENT STAGE

LMA790-3-LM
APOLLO OPERATIONS HANDBOOK
SUBSYSTEMS DATA



300LM4-156

Figure 2.1-10. TTCA Manipulations



300LMA-52

Figure 2.1-9. ACA Manipulations

2.1.1.3.2 Thrust/Translation Controller Assemblies. (See figure 2.1-10.)

The TTCA's control LM translation in any axis; they are functionally integrated translation and thrust controllers. The astronauts use these assemblies to command vehicle translations by firing RCS thrusters and to throttle the descent engine between 10% and 92.5% thrust magnitude. The controllers are three-axis, T-handle, left-hand controllers, mounted with their longitudinal axis approximately 45° from a line parallel to the LM Z-axis (forward axis).

A lever on the right side of the TTCA enables the astronaut to select either of two control functions: (1) translation control in the Y-axis and Z-axis, using the RCS thrusters, and descent engine throttling to control X-axis translation and (2) translation control in all three axes, using the RCS thrusters. Due to the TTCA mounting position, vehicle translations correspond to astronaut hand movements when operating the controller. Moving the T-handle to the left or right commands translation along the Y-axis. Moving the T-handle inward or outward commands translation along the Z-axis. Moving the T-handle upward or downward commands translation along the X-axis, using the RCS thrusters, when the select lever is in the down position. When the lever is in the up position, upward or downward movement of the TTCA increases or decreases, respectively, the magnitude of descent engine thrust. Regardless of the select lever position selected, the TTCA can command translation along the Y-axis and Z-axis, using the RCS thrusters. ■

2.1.1.3.3 Attitude and Translation Control Assembly.

The ATCA controls vehicle attitude and translation by issuing on-off commands to the RCS thrusters. In primary guidance control, attitude and translation commands are generated by the LGC and applied directly to jet drivers within the assembly. In the abort guidance path, the ATCA receives translation commands from a TTCA, rate-damping signals from the RGA, and attitude rate commands and pulse commands from the ACA. The ATCA combines attitude and translation commands in its logic network to select the proper thruster to be fired for the desired translation and/or rotation.

LMA790-3-LM
APOLLO OPERATIONS HANDBOOK
SUBSYSTEMS DATA

The guidance computational section provides trajectory computation and selection, steering computation, and midcourse-correction computation. This computational section receives data relating to the CSM state vector and the LM state vector from the LGC in other external source through the AGS input selector logic. The state vector is the vehicles attitude and velocity for a given mission time. Body-referenced steering errors are received from the stabilization and alignment computational section, for trajectory computation. The LM abort guidance problem consists of solving the equations of the selected guidance maneuver, including steering, attitude, and engine control computations. Outputs of the guidance computational section, through the output select logic circuits, include engine on and off signals to the CES, and velocity to be gained (selectable by DEDA readout).

2.1.1.3 Control Electronics Section. (See figure 2.1-8.)

The CES processes attitude and translation signals when operating in the primary guidance mode or the abort guidance mode.

When operating in the primary mode, the CES converts RCS commands to the required electrical power to operate the RCS solenoid valves. The CES accepts discrete (on and off) descent engine gimbal commands and, upon receipt of an on command, causes the descent engine to move about its gimbal axis. The CES accepts LGC and manual engine on and off commands and routes them to the MPS to fire or stop the descent or ascent engine. The CES accepts LGC and manual thrust commands to throttle the descent engine (10% to 92.5% of maximum thrust). The CES also provides manual attitude and translation commands to the LGC.

When operating in the abort guidance mode, the CES accepts attitude error signals from the AGS, or manual attitude rate commands from the attitude hand controller or rate-damping signals from a gyro assembly, and fires the RCS thrusters to achieve attitude control. The CES accepts manual translation commands and fires the appropriate thrusters to accelerate the LM in the desired direction. The CES automatically gimbals the descent engine for trim control in accordance with signal polarity. The CES accepts AGS and manual engine on and off commands and routes them to the descent or ascent engine. The CES accepts manual throttle commands to control descent engine thrust and accepts manual rotational, low-amplitude acceleration, open-loop commands.

The CES comprises two attitude controller assemblies (ACA's), two thrust/translation controller assemblies (TTCA's), an attitude and translation control assembly (ATCA), a rate gyro assembly (RGA), descent engine control assembly (DECA), two gimbal drive actuators (GDA's), an ascent engine arming assembly (AEAA), and three stabilization and control (S&C) control assemblies.

2.1.1.3.1 Attitude Controller Assemblies. (See figure 2.1-9.)

The ACA's are right-hand pistol grip controllers, which the astronauts use to command changes in vehicle attitude. Each ACA is installed with its longitudinal axis approximately parallel to the X-axis. Each ACA supplies attitude rate commands proportional to the displacement of its handle, to the LGC and the ATCA; an out-of-detent discrete each time the handle is out of its neutral position; and a followup discrete to the AGS each time the controller is out of detent. A trigger-type push-to-talk switch on the pistol grip handle of the ACA is used for communication with the CSM and ground facilities.

As the astronaut uses his ACA, his hand movements are analogous to vehicle rotations. Clockwise or counterclockwise rotation of the controller commands yaw right or yaw left, respectively. Forward or aft movement of the controller commands vehicle pitch down or up, respectively. Left or right movement of the controller commands roll left or right, respectively.

The ACA's are also used in an incremental landing point designator (LPD) mode, which is available to the astronauts during the final approach phase. In this mode, the angular error between the designated landing site and the desired landing site is nulled by repetitive manipulation of an ACA. LPD signals from the ACA are directed to the LGC, which issues commands to move the designated landing site incrementally along the Y-axis and Z-axis.

The ASA consists of three strapped-down pendulous accelerometers, three strapped-down gyros, and associated electronic circuitry. The accelerometers and gyros (one each for each vehicle axis) sense body-axis motion with respect to inertial space. The accelerometers sense acceleration along the vehicle orthogonal axis. The gyros and accelerometers are securely fastened to the vehicle X, Y, and Z axes so that motion along or around one or more axis is sensed by one or more gyros or accelerometers.

2.1.1.2.2 Data Entry and Display Assembly. (See figure 2.1-7.)

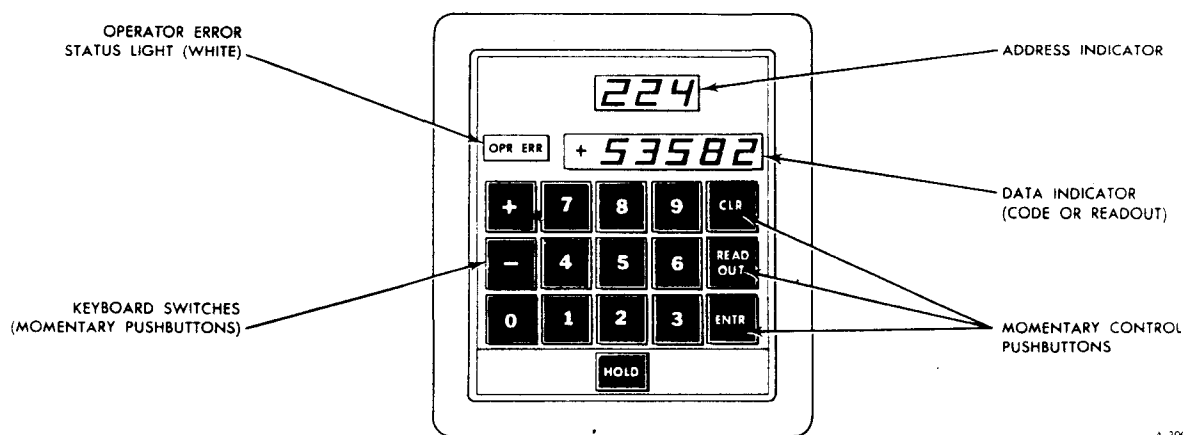
The DEDA (panel 6) is used by the astronauts to select the desired mode of operation, insert the desired targeting parameters, and monitor related data throughout the mission. Essentially, the DEDA consists of a control panel to which electroluminescent displays and data entry pushbuttons are mounted and a logic enclosure that houses logic and input-output circuits.

2.1.1.2.3 Abort Electronics Assembly.

The AEA is a general-purpose, high-speed, 4,096-word digital computer that performs basic strapped-down guidance system calculations and the abort guidance and navigation steering calculations. The computer uses a fractional two's complement, parallel arithmetic section, and parallel data transfer. The AEA has three software computational sections: stabilization and alignment, navigation, and guidance.

The stabilization and alignment computational section computes stabilization and alignment on generation of mode signals by the DEDA. These mode signals (attitude hold, guidance steering, Z-axis steering, PGNS-to-AGS alignment, lunar align, gyro and accelerometer calibration, and body-axis align) determine the operation of the stabilization and alignment computational section in conjunction with the navigation and guidance computational sections. The body-referenced steering error signals and total attitude sine and cosine signals are used to control the FDAI. Direction cosine data are routed to the navigation computational section, where they are used in computing lateral velocity and inertial acceleration data.

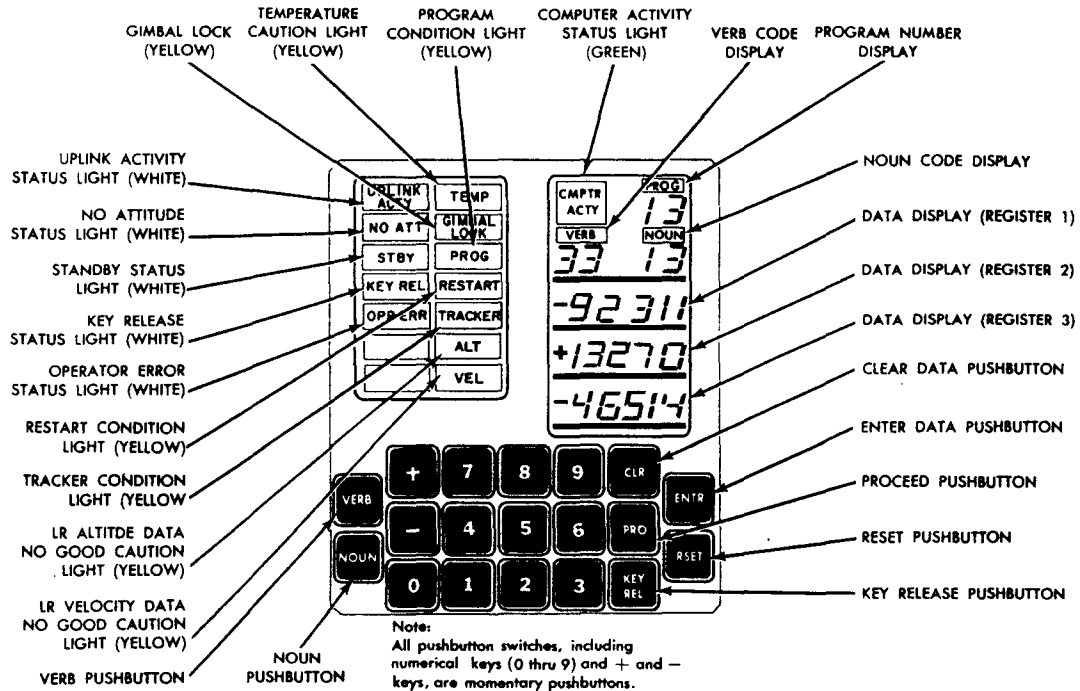
The navigation computational section uses accelerometer inputs received from the ASA, via AEA input logic circuits, to calculate LM position and velocity in the inertial reference frame. The navigation computational section supplies total velocity, altitude, and altitude-rate data, and lateral velocity data in the LM reference frame, to the output logic circuits. Velocity data are routed to the DEDA, altitude-rate data are routed to the ALT RATE indicator (panel 1), and lateral velocity data are routed to the X-pointer indicators (panels 1 and 2). Velocity and position data are routed to the guidance computational section, for computing LM orbital parameters.



A 300LM4 254

Figure 2.1-7. Data Entry and Display Assembly - Pictorial

GUIDANCE, NAVIGATION, AND CONTROL SUBSYSTEM



300LMS-18

Figure 2.1-5. Display and Keyboard Assembly

The AGS provides steering commands for three steering submodes: attitude hold, guidance steering, and acquisition steering. The attitude hold submode maintains the vehicle attitude that exists when the submode is entered. In the guidance steering submode, the AEA generates attitude commands to orient the LM X-axis so that it lies along the direction of the thrust vector. In the acquisition steering submode, the AEA generates attitude commands to orient the LM Z-axis along the estimated line of sight (LOS) between the LM and CSM.

The AGS outputs an engine-on or engine-off command during all thrusting maneuvers. If the PGNS is in control, the command is a followup of the signal produced by the PGNS. If the AGS is in control, the engine-on command can be routed only after the appropriate switches are set and the ullage maneuver has been performed. When proper velocity-to-be-gained are achieved, an engine-off command is issued.

The AGS uses RR angle information and accepts range and range-rate information from the RR for updating LM navigation so that the LM Z-axis is toward the CSM, or for midcourse correction. These data are manually inserted into the AEA by the astronaut by using the DEDA.

The AGS automatically aligns the strapped-down inertial system of the ASA by computing the direction cosines that relate the LM body axes to the desired inertial coordinate system. It also provides in-flight gyro and accelerometer calibration to compensate for fixed non-g gyro drift, and telemetry data for MSFN through the IS.

2.1.1.2.1 Abort Sensor Assembly.

The ASA, by means of gyros and accelerometers, provides incremental attitude information around the vehicle X, Y, and Z axes and incremental velocity changes along the vehicle X, Y, and Z axes. Data pulses are routed to the AEA, which uses the attitude and velocity data for computation of steering errors. The ASA is mounted on the navigation base above the astronauts' heads, between the crew compartment and the thermal and micrometeoroid shield.

GUIDANCE, NAVIGATION, AND CONTROL SUBSYSTEM

Mission LM Basic Date 1 February 1970 Change Date 15 June 1970 Page 2.1-9

In the fixed memory, addresses are assigned to instruction words to specify the sequence in which they are to be executed; blocks of addresses are reserved for data, such as constants and tables. Information is placed into the fixed memory permanently by weaving patterns through the magnetic cores. Information is written into assigned locations in the erasable memory with the DSKY, uplink, or program operation.

Both memories use a common address register (S-register) and an address decoder in the central processor. When the S-register contains an address pertaining to the erasable memory, the erasable memory cycle timing is energized. Pulses sent to the erasable memory cycle timing then produce strobe signals for the read, write, and sense functions. The erasable memory selection logic receives an address and a decoded address from the central processor and produces selection signals, which permit data to be written into, or read out from, a selected storage location. When a word is read out from a storage location in the erasable memory, the location is cleared. A word written into the erasable memory, through the memory buffer register in the central processor, by a write strobe operation. A word read from a storage location is applied to the amplifiers. The amplifiers are strobed and the information is entered into a buffer register of the central processor. The memory buffer register receives information from both memories.

The address in the S-register energizes the fixed memory cycle timing when a location in the fixed memory is addressed. Pulses sent to the fixed memory cycle timing produce the strobe signals for the read and sense functions. The selection logic receives an address from the write lines and a decoded address from the S-register, and produces selection signals for the core rope. The content of a storage location in the fixed memory is strobed from the fixed memory sense amplifiers to the erasable memory sense amplifiers and then entered into the memory buffer register of the central processor.

Alarm-Detection Network. The alarm-detection network consists of temperature, voltage, scaler, double frequency scaler oscillator, memory clamping, and the warning filter and integrator circuits. The alarm-detection network monitors LGC operation. If an LGC failure is detected, a failure signal is routed to the DSKY for display. An LGC power failure is also displayed by the LGC warning light (panel 1).

2.1.4.3.2 Display and Keyboard Assembly. (See figure 2.1-33.)

The DSKY consists of a keyboard, display panel, condition indicators, and a relay package. The keyboard enables the astronauts to insert data into the LGC and to initiate LGC operations. Through the keyboard, the astronauts can also control ISS moding. The DSKY display panel provides visual indications of data being loaded into the LGC, LGC condition, and LGC program. The display panel also provides the LGC with a means of displaying or requesting data. The condition indicators display PGNS status and malfunctions. The controls and displays associated with the DSKY are discussed in section 3.

Keyboard. The DSKY keyboard is used to insert or read out LGC data. The keyboard consists of 10 numerical pushbuttons (0 to 9), two sign pushbuttons (+ and -), and seven instruction pushbuttons (ENTR, CLR, VERB, NOUN, RSET, PRO, and KEY REL). All the pushbuttons, except the PRO pushbutton, have five-bit codes associated with them; they convey information to the LGC. The PRO pushbutton is hardwired into the LGC power supplies.

Displays and Indicators. There are two types of displays on the DSKY: control displays and data displays. Each display can display any decimal character or remain blank. The indicators on the DSKY are referred to as condition lights; they represent various PGNS operating conditions. Each control display (VERB, NOUN, and PROG) can display two decimal characters or remain blank. The VERB and NOUN displays can also flash. The data displays are three separate registers, referred to as R1, R2, and R3. Each register can display as many as five decimal characters, with or without a plus or minus sign, or remain blank. Each of the 11 condition lights on the DSKY is labeled with the PGNS condition it represents; it goes on if that condition occurs. The condition lights and the conditions they represent are described in section 3.

GUIDANCE, NAVIGATION, AND CONTROL SUBSYSTEM

Mission LM Basic Date 1 February 1970 Change Date _____ Page 2.1-77

SECTION 3
CONTROLS AND DISPLAYS

INTRODUCTION

This section contains detailed descriptions of the LM controls and displays. A cross-reference system, which permits quick and easy reference to any control or display, is provided in table 3-1. This table lists each item alphabetically, the item reference designation, the panel that contains the item, and the page number on which the item is to be found.

Figures 3-1 and 3-2, at the end of this section, show the controls and displays, panel areas, and panel numbers. ■

Column headings for the tabular presentation, and the description of the entries made under each, are as follows:

- Location/Ref Des. Notes the location, by panel number, of a particular control or display in the LM, and lists the reference designation.
- Name and Position. Gives the placarded nomenclature of the control or display and the control positions exactly as placarded on the panel. In the absence of placards, a functional name is assigned, and control positions are described.
- Function. Gives a brief description of the function of the control or display.
- Circuit Breaker. Gives the nomenclature and location of the circuit breaker(s) that controls the electrical power to the control or display.
- Power Source. Gives the rating and nomenclature of the immediate bus or source that supplies power to the control or display.
- Remarks. Contains additional data and explanatory remarks.

Location/ Ref Des	Name and Position	Function	Circuit Breaker	Power Source	Remarks
Panel 1 (cont)					
9M12	MISSION TIMER indicator	Displays digital readout of elapsed time in hours, minutes, and seconds. Indicator is controlled with TIMER CONT and SLEW CONT switches on panel 5. Maximum time displayed is 999 hours 59 minutes 59 seconds.	FLIGHT DISPLAYS: MISSION TIMER (panel 11)	CDR's 28-vdc bus	MISSION TIMER indicator can only count up. If PCMTTE timing pulses fail, an internal tuning fork supplies the 10 pps clock frequency, and a display appears before the first indicator digit.
9M11	EVENT TIMER indicator	Displays digital readout of time (0 minutes 0 seconds to 59 minutes 59 seconds). EVENT TIMER is controlled with RESET/COUNT, TIMER CONT, SLEW CONT MIN, and SLEW CONT SEC switches on panel 3.	FLT DISP: EVENT TIMER/SE FDAI (panel 16)	LMP's 28-vdc bus	EVENT TIMER indicator can count up or down. If it fails, MISSION TIMER indicator can be used.
9M2	X-Pointer indicator	Displays forward and lateral velocities, lateral velocities only, or rendezvous radar LOS elevation and azimuth angular rates, depending upon setting of RATE/ERR MON switch and MODE SEL switch.	FLIGHT DISPLAYS: CDR X-PNTR (panel 11)	CDR's 28-vdc bus	Forward and lateral velocities are coincident with LM Z and Y body axis velocities when PGNS drives display. When LR drives display, forward and lateral velocities are coincident with LM Z and Y axis (from hi-gate point to touchdown). When AGS drives display, lateral Y-axis velocity only is displayed. When RR is selected, LOS rates are displayed.
16DS91	X-Pointer indicator power failure indicator	Red light goes on when power to indicator is interrupted.	N/A	N/A	
1DS1	LUNAR CONTACT light	Light goes on when landing gear probe touches surface, to indicate that engine should be turned off.	N/A	N/A	Light goes off when either engine stop switch is pressed.

LM A790-3-LM
APOLLO OPERATIONS HANDBOOK
CONTROLS AND DISPLAYS

Location/ Ref Des	Name and Position	Function	Circuit Breaker	Power Source	Remarks
Panel 1 (cont) 9S17	X-POINTER SCALE switch	Controls scale range of X-pointer indicator.	N/A	N/A	
	HI MULT LO MULT	When LOS azimuth and elevation rates are displayed, this position provides a scale of ± 20 mrad/sec. When horizontal velocities are displayed, this position illuminates the X10 multiplier to provide range of ± 200 fps. When LOS azimuth and elevation rates are displayed, this position illuminates the X.1 multiplier to provide a range of ± 2 mrad/sec. When horizontal velocities are displayed, this position provides range of ± 20 fps.			
9M7	THRUST indicator	Displays descent engine chamber pressure, which corresponds to thrust on ENG scale (left pointer) and manual or LGC thrust commands to the engine on CMD scale (right pointer). Both scales read from 0% to 100%. Automatic or manual throttle commands can be displayed, depending on setting of THR CONT switch. The CMD (right) scale indicates 10% thrust command, even when engine is not firing because indicator input is not the actual thrust command input to engine unless the engine is on, and TTCA is at minimum position. Thrust command to engine, in AUTO mode, (GUID CONT switch is set to PGNS) is sum of thrust commands from LGC and TTCA. TTCA provides a minimum thrust	FLIGHT DISPLAYS: THRUST (panel 11)	CDR's 28-vdc bus	Normally, during LGC command authority, both pointers will be aligned. If not, a malfunction exists or manual throttle authority is being introduced to enable

Location/ Ref Des	Name and Position	Function	Circuit Breaker	Power Source	Remarks
Panel 1 (cont)		command of 10% at all times; it cannot be set to zero. When THR CONT switch is set to AUTO position and TTCA is at minimum position, LGC commands 10% less than required and is summed with 10% command from TTCA to provide required thrust level. When THR CONT switch is set to MAN, LGC commands are removed and all thrust commands originate from TTCA.			smooth transition to full manual control. If manual control is desired, THR CONT switch is set to MAN when CMD pointer reaches 10%; pointers will then realign. If TTCA control is moved beyond minimum thrust position, LGC commands correspondingly less thrust. It is, therefore, possible in AUTO mode to command more thrust than LGC requires, but not possible to command less.
16DS92	THRUST indicator power failure indicator	Red light goes on when power to indicator is interrupted.	N/A	N/A	
	MAIN PROPULSION QUANTITY	Indicate quantity of propellant remaining in descent tanks.	AC BUS B: He/PQGS PROPUL DISP (panel 11)	CDR's 115-vac bus B	Quantity indicator is operative from 0% to 95%.
10M5A	OXIDIZER	Displays digital readout of percentage (0% to 95%) of usable oxidizer remaining in descent tank selected with PRPLNT QTY MON switch.			
10M5B	FUEL	Displays digital readout of percentage (0% to 95%) of usable fuel remaining in descent tank selected with PRPLNT QTY MON switch.			
10M1	MAIN PROPULSION HELIUM indicator	Displays digital readout of supercritical helium pressure in descent cryogenic storage vessel, pressure in descent ambient helium bottle, and temperature or pressure	AC BUS B: He/PQGS PROPUL DISP (panel 11)	CDR's 115-vac bus B	

Location/ Ref Des	Name and Position	Function	Circuit Breaker	Power Source	Remarks
Panel 1 (cont)		in ascent helium tanks, as selected with HELIUM MON switch.			
10M4	TEMP indicator	Indicates fuel and oxidizer bulk temperature (+40° to +100° F) in ascent or descent propellant tanks, as selected with PRPLNT TEMP/PRESS MON switch.	PROPUL: DISP/ENG OVRD LOGIC (panel 16)	LMP's 28-vdc bus	
10M3	PRESS indicator	Indicates fuel and oxidizer pressures (0 to 300 psia) for ascent or descent propellant sections, as selected with PRPLNT TEMP/PRESS MON switch.	PROPUL: DISP/ENG OVRD LOGIC (panel 16)	LMP's 28-vdc bus	
16DS90	PRESS indicator power failure indicator	Red light goes on when power to indicator is interrupted.	N/A	N/A	
6S2	MASTER ALARM pushbutton/light Audible tone	Light flashes when any caution or warning light goes on, except C/W PWR caution and DES QTY warning lights. Initiated in conjunction with MASTER ALARM pushbutton-light and caution/warning array light(s).	LTG: MASTER ALARM (panel 16)	LMP's 28-vdc bus	Pressing Commander's or LM Pilots switch-light extinguishes both MASTER ALARM lights and terminates audible tone, but has no effect on caution or warning light(s).
9S2	RATE/ERR MON switch RNDZ RADAR	Selects inputs for FDAI and X-pointer indicator. Shaft and trunnion angles from the RR are displayed by pitch and yaw error needles of FDAI; LOS azimuth and elevation rates are displayed on X-Pointer indicator (illuminating LOS AZ and LOS ELEV).	N/A	N/A	When switch is set to LDG RDR/CMPTR, data input is dependent upon settings of MODE SEL and ATTITUDE MON switches.

Page 3-20 Mission LM Basic Date 1 February 1970 Change Date

PANEL 1

Location, Ref Des	Name and Position	Function	Circuit Breaker	Power Source	Remarks
Panel 1 (cont)	LDG RDR/CMPTTR	Vehicle attitude errors (PGNS or AGS as selected with ATTITUDE MON switch) are displayed by FDAI error needles. Forward and lateral velocities from the LR (PGNS or AGS as selected with MODE SEL switch) are displayed on the X-pointer indicator (illuminating LAT VEL and FWD VEL).			AGS provides only lateral velocity data.
9S3	ATTITUDE MON switch	Selects input for FDAI total attitude display and input for attitude error needles during landing.	N/A	N/A	
	PGNS	LM total attitude signals, after conditioning by GASTA, are fed to FDAI ball; LGC attitude error signals are fed to FDAI pitch, roll, and yaw error needles.			
	AGS	AGS LM total attitude signals are fed to FDAI ball; AGS attitude error signals are fed to FDAI pitch, roll, and yaw error needles.			
9M1	Flight director attitude indicator (FDAI)	Displays total vehicle attitudes, attitude rates and attitude errors, or vehicle attitude, attitude rates, and RR shaft and trunnion angles, depending on setting of RATE/ERR MON switch. Setting ATTITUDE MON switch selects PGNS or AGS as source of vehicle total attitude and attitude errors displayed on FDAI. Shaft and trunnion angles are displayed by pitch and yaw error needles, respectively, when RATE/ERR MON switch is set to RNDZ RADAR.	AC BUS A: CDR FDAI and FLIGHT DISPLAYS: CDR FDAI (panel 11)	CDR's 115-vac bus A CDR's 28-vdc bus	Attitude rates displayed are always obtained from CES rate gyro assembly.

Location/ Ref Des	Name and Position	Function	Circuit Breaker	Power Source	Remarks
Panel 1 (cont) 9M9 9M4 9S6	Altitude/range indicator	Displays range/range rate data or altitude/altitude rate data as selected with RNG/ALT MON switch. Range/range rate data are from RR. Altitude/altitude rate data are from LR, PGNS, or AGS, as selected with MODE SEL switch. When LR data is selected, true altitude and altitude rate data are available from low-gate point to touchdown if LM X-axis is vertical. Before reaching low-gate point, only true altitude data are available from LR. When PGNS or AGS is selected, inertially derived altitude and altitude rate data are available for display.	FLIGHT DISPLAYS: RNG/RNG RT ALT/ALT RT and AC BUS A; RNG/RNG RT ALT/ALT RT (panel 11)	CDR's 28-vdc bus A	
	T/W indicator	Displays instantaneous X-axis acceleration in lunar g units (1 lunar g = 5.23 ft/sec ²).	N/A	N/A	Indicator, a self-contained accelerometer, may be used to provide a gross check of engine performance, because any given throttle setting provides specific acceleration when the vehicle has given mass.
	GUID CONT switch PGNS	Selects PGNS or AGS for guidance control mode of LM. Provides 800-cps power to activate ACA and TTCA, proportional rate command inputs to LGC from ACA, LGC engine on-off signals, descent engine gimbal trim command outputs from LGC, translation on-off commands from TTCA, enabling signals for primary preamplifiers of ATCA, and applies follow-up signal to AEA.	N/A	N/A	Switch is spring loaded to center (lock) position. When AGS mode is selected, PGNS remains operational. PGNS data may be displayed if desired.

Location/ Ref Des	Name and Position	Function	Circuit Breaker	Power Source	Remarks
Panel 1 (cont)	AGS	Provides 800-cps power to activate ACA and TTCA, proportional rate commands to ATCA from ACA, AGS mode trim command, and enabling signals to the abort preamplifiers of ATCA.			
	9S9 LDG RADAR PGNS AGS	Selects radar or computer data for display on X-Pointer and altitude/range indicators. Radar altitude, altitude rate, and forward and lateral velocity are displayed. LGC-computed altitude, altitude rate, and forward and lateral velocity are displayed. AGS-computed altitude, altitude rate and lateral velocity are displayed.	N/A	N/A	Data from selected source are displayed on appropriate horizontal velocity indicator only when RATE/ERR MON switches are set to LDG RDR/COMPTR. Data from selected source are displayed on altitude/range indicators only when RNG/ALT MON switch is set to ALT/ALT RT.
	9S4 RNG/ALT MON switch RNG/RNG RT	Selects display legend on altitude/range indicator and displays RR range/range rate data or LR or computer altitude/altitude rate data. RR range and range rate data are displayed on altitude/range indicator.	N/A	N/A	

LM1790-3-LM
APOLLO OPERATIONS HANDBOOK
CONTROLS AND DISPLAYS

Location/ Ref Des	Name and Position	Function	Circuit Breaker	Power Source	Remarks
Panel 1 (cont)	ALT/ALT RT	Data from source selected with MODE SEL switch are displayed on altitude/range indicator.			
988	SHFT/TRUN ✕ switch +50° ±5°	Selects range of RR shaft and trunnion angles to be displayed by FDAI pitch and yaw error needles when either RATE/ERR MON switch is set to RNDZ RADAR. Full deflection of FDAI pitch and yaw error needles indicates shaft and trunnion angles of +50° or -50° or greater. Full deflection of FDAI pitch and yaw error needles indicates shaft and trunnion angles of +5° or -5° or greater.	FLIGHT DISPLAYS: CDR FDAI (panel 11)	CDR's 28-vdc bus	Less-than-full-deflection positions of needles are linearly proportional to the sine of the shaft and trunnion angles less than +50° or -50°. Less-than-full-deflection positions of needles are linearly proportional to shaft and trunnion angles less than +5° or -5°.
9S29	RATE SCALE switch 25°/sec 5°/sec	Selects scale factors of roll, pitch, and yaw rate scales of FDAI. Selects 25° per second full scale for rate scales. Selects 5° per second full scale for rate scales.	N/A	N/A	
1S25	ACA PROP switch ENABLE DISABLE	Enables isolation of 28 volts, 800 cps to ACA transducer primary coil. Enables normal operation of ACA in proportional rate mode. Disables ACA proportional rate command capability.	N/A	N/A	If ACA is disabled, only the landing mission phase is affected.

Mission LM Basic Date 1 February 1970 Change Date
PANEL 1
Page 3-23

PANEL
1

Page 3-24 Mission LM Basic Date 1 February 1970 Change Date 15 June 1970
PANEL 1

Location/ Ref Des	Name and Position	Function	Circuit Breaker	Power Source	Remarks
Panel 1 (cont)	ENG THRUST CONT switches				
	1S11 THR CONT switch	Permits switching from automatic (LGC) throttle control to manual throttle control.			
	AUTO	LGC command signals are summed with manual command signals from TTCA selected with MAN THROT switch for descent engine throttle control. TTCA always provides minimum 10% command; it cannot be set below this level.			When switch is set to AUTO, THRUST indicator displays LGC command plus 10% fixed bias. Manual throttle commands may be introduced by displacing active TTCA. This causes displayed % THRUST CMD to decrease because only LGC commands plus fixed bias are displayed.
	MAN	Interrupts LGC throttle commands insuring that descent engine throttle is fed only by manual throttle command signals.	STAB/CONT; ENG CONT (panel 11)	CDR's 28-vdc bus	CMD THRUST indicator displays manual throttle commands.
1S10	MAN THROT switch	Selects thrust/translation controller for manual adjustment of descent engine thrust level, if corresponding THROTTLE-JETS control select level is set to THROTTLE. This switch also routes manual throttle commands from controller to DECA.	N/A	N/A	
	CDR	Enables Commander's thrust/translation controller to adjust descent engine thrust level.			
	SE	Enables LM Pilot's thrust/translation controller to adjust descent engine thrust level.			

Location/ Ref Des	Name and Position	Function	Circuit Breaker	Power Source	Remarks
Panel 1 (cont)	1S1	ENG ARM switch	STAB/CONT: ENG ARM (panel 16)	LMP's 28-vdc bus	Switch is spring loaded to center (lock) position. Regardless of switch setting, appropriate engine is armed if ABORT or ABORT STAGE switches are actuated.
		ASC	STAB/CONT: ENG CONT and STAB/CONT: ENG START OVRD (panel 11)	CDR's 28-vdc bus	
		OFF			
		DES			
	1S6	ATT/TRANSL switch	STAB/CONT: ATT DIR CONT (panel 11)	CDR's 28-vdc bus	This control can only be used with AGS.
		4 JETS			
		2 JETS			
1S9	BAL CPL switch	Selects either balanced pairs of RCS jets in a couple or unbalanced X-axis RCS jets to maintain pitch and roll attitude during ascent engine thrust phase when AGS is in guidance control loop.	N/A	N/A	Used only in conjunction with AGS.
	ON	Applies 28-vdc enable voltage to abort preamplifiers 1, 5, 9, 13.			Normally in this position (balanced couples) during initial phases of ascent, for maximum stabilization and control of any center-of-gravity thrust vector misalignment.

Location/ Ref Des	Name and Position	Function	Circuit Breaker	Power Source	Remarks
Panel 1 (cont)	OFF	Removes enable voltage from abort preamplifiers 1, 5, 9, 13.			After some minimum burn time when balanced-couple operation is not required, switch can be set to OFF to conserve fuel.
10S3	ASCENT He REG 1 switch	Selects inputs to normally open, latching solenoid-operated helium valve upstream of primary pres- sure regulator.	PROPUL: ASC He REG (panel 16)	LMP's 28-vdc bus	Switch is spring loaded to center position.
	OPEN	Energizes solenoid that drives helium valve to open position.			Gray talkback is displayed when valve is open.
	Center	Removes solenoid excitation; valve remains in last commanded position.			
	CLOSE	Energizes solenoid that drives helium valve to closed position.			Barber-pole talkback is displayed when valve is closed.
10FL1	ASCENT He REG 1 talkback		PROPUL: DISP/ENG OVRD LOGIC (panel 16)	LMP's 28-vdc bus	
	Gray display	Indicates helium primary line solenoid valve controlled by ASCENT He REG 1 switch is open.			
	Barber-pole display	Indicates helium primary line solenoid valve controlled by ASCENT He REG 1 switch is closed.			
10S4	ASCENT He REG 2 switch	Selects inputs to normally open latching solenoid-operated helium valve upstream of secondary pres- sure regulator.	PROPUL: ASC He REG (panel 16)	LMP's 28-vdc	Switch is spring loaded to center position.
	OPEN	Energizes solenoid that drives helium valve to open position.			Gray talkback is displayed when valve is open.

Location/ Ref Des	Name and Position	Function	Circuit Breaker	Power Source	Remarks
Panel 1 (cont)	Center	Removes solenoid excitation; valve remains in last commanded position.			
	CLOSE	Energizes solenoid that drives helium valve to closed position.			Barber-pole talkback is displayed when valve is closed.
	10FL2	ASCENT He REG 2 talkback			
	Gray display	Indicates helium secondary line solenoid valve controlled by ASCENT He REG 2 switch is open.	PROPUL: DISP/ENG OVRD LOGIC (panel 16)	LMP's 28-vdc bus	
	Barber-pole display	Indicates helium secondary line solenoid valve controlled by ASCENT He REG 2 switch is closed.			
	10S5	DESCENT He REG 1 switch			
	OPEN	Selects inputs to normally open, latching solenoid-operated helium valve upstream of primary pres- sure regulator.	PROPUL: DES He REG-VENT' (panel 11)	CDR's 28-vdc bus	Switch is spring loaded to center position.
10FL3	Center	Energizes solenoid that drives helium valve to open position.			Gray talkback is displayed when valve is open.
	CLOSE	Removes solenoid excitation; valve remains in last commanded position.			
		Energizes solenoid that drives helium valve to closed position.			Barber-pole talkback is displayed when valve is closed.
	DESCENT He REG 1 talkback				
	Gray display	Indicates helium primary line solenoid valve controlled by DESCENT He REG 1 switch is open.	PROPUL: DISP/ENG OVRD LOGIC (panel 16)	LMP's 28-vdc bus	

Mission LM Basic Date 1 February 1970 Change Date Page 3-59

Location/ Ref Des	Name and Position	Function	Circuit Breaker	Power Source	Remarks
Panel 3 1S8	ENG GMBL switch	Enables or disables engine gimbal drive capability.	AC BUS A: DECA GMBL (panel 11)	CDR's 115-vac bus A	Normally set to ENABLE. If ENG GMBL caution light goes on, astronaut must set switch to OFF.
	ENABLE	Gimbal trim signals are enabled to displace descent engine along Y and Z axes.			
	OFF	Power to the GDA's is removed, disabling the gimbal drive capability.			
1S19	DES ENG CMD OVRD switch	Supplies alternate source of 28-vdc to descent engine shut-off valves.	STAB/CONT: ABORT STAGE (panel 11)	CDR's 28-vdc bus	Switch is utilized to keep valves open if DECA power fails.
	ON	Applies 28 vdc to descent engine shut-off valves.			
17S5	LDG ANT switch		N/A	N/A	Because antenna cannot be driven to its first position by LGC and it is necessary to check out completely two-position capability and provide override of position drive signal if LR antenna interface fails, DES and HOVER positions are used.
	AUTO	Enables LGC to position LR antenna as function of mission phase.			
	DES	Drives antenna to first position used before hover point.			
	HOVER	Drives antenna to its second position for final landing phase.			

Location/ Ref Des	Name and Position	Function	Circuit Breaker	Power Source	Remarks
Panel 3 (cont) 17S1	SLEW switch	Slews trunnion or shaft servos, or both, positioning RR antenna for manual acquisition of target and subsequent automatic tracking.	N/A	N/A	Two antenna slew rates are available using SLEW RATE switch. Target acquisition is determined from SIGNAL STRENGTH indicator and auto track mode is enabled with rendezvous radar mode selector switch. RR antenna alignment along LM axis can be determined from FDAI pitch and yaw error needles when RATE/ERR MON switch is set to RNDZ RADAR.
	UP	Slews shaft servos to up position, positioning RR antenna for manual acquisition of target and subsequent automatic tracking.			
	RIGHT	Same as for UP position, except that slewing is to right position.			
	DOWN	Same as for UP position, except that slewing is to down position.			
	LEFT	Same as for UP position, except that slewing is to left position.			
	RENDEZVOUS RADAR mode selector switch	Enables control of RR antenna drive.	N/A	N/A	Once target is acquired, radar automatically maintains track and NO TRACK light goes off.
	AUTO TRACK	Maintains track by comparing received radar signals from three channels; errors are used to drive radar servos.			
	SLEW	Enables manual control of RR antenna drive with SLEW switch.			
	LGC	LGC automatically drives antenna towards target.			
	17S3				

Location/ Ref Des	Name and Position	Function	Circuit Breaker	Power Source	Remarks
Panel 3 (cont) 11S4 11S3 11S5	ATTITUDE CONTROL ROLL, PITCH, and YAW switches MODE CONT PULSE DIR	Used in conjunction with MODE CONTROL switches to establish attitude control in roll, pitch, and yaw. Normally these switches are set to MODE CONT while PGNS is in control. Enables use of proportional mode of ACA when MODE control selector switch is set to ATT HOLD and PGNS is controlling. When MODE CONTROL selector switch is set to AUTO, LGC supplies attitude commands to maneuver LM. Under AGS control, function of MODE CONT positions are the same manner as when PGNS is controlling. Under PGNS control this position is inoperative. Under AGS control, the PULSE and DIR positions differ only in how RCS thrusters fire. When PULSE is selected, two jets in selected axis fire at constant pulsed rate as long as ACA is displaced more than one-fourth of full throw. PULSE and DIR positions provide open loop acceleration. Rates produced must be nulled to zero by inducing opposite acceleration command. Attitude is not maintained automatically until switch is returned to MODE CONT. Under PGNS control, when MODE CONTROL selector switch is set to ATT HOLD or AUTO, DIR mode can be used in any axis. Each switch must be set to DIR independently. Displacing ACA one-fourth of full	N/A	N/A	

Location/ Ref Des	Name and Position	Function	Circuit Breaker	Power Source	Remarks
Panel 3 (cont)		throw causes continuous two-jet firing for selected axes. As in PULSE position, open loop acceleration is provided; it must be nulled in as during pulse operation.			
11S24	PGNS MODE CONTROL switch	Affects outputs of ACA's, TTCA's and LGC.	N/A	N/A	Switch is associated with ATTITUDE CONTROL switches and GUID CONT switch.
	OFF	Establishes the following conditions: (a) Limits operation of the ACA's to the hardover position. (b) Enables two-get direct attitude control in axis, selected with ATTITUDE CONTROL switches. (c) Disables translation outputs of TTCA's and LGC attitude commands.			TTCA throttle commands are not affected. LGC engine-off signals are not enabled; engine must be manually turned off.
	ATT HOLD	In this mode, astronauts can command angular rates proportional to displacement of ACA's. Mode is functional with either PGNS or AGS controlling operation. When ACA is out of detent, ATTITUDE CONTROL switches set to MODE CONT, and LGC not in control, attitude loops are opened and axis transducers are compared to rate gyro outputs to provide proportional commands. While maneuvering, attitude synchronizers follow as attitude changes. If PGNS control is selected, astronauts can command attitude rates proportional to ACA displacement, and attitude is held when ACA is			When ATTITUDE CONTROL switches are set to PULSE or DIR, following attitude command capabilities result: With GUID CONT switch set to PGNS and one or more ATTITUDE CONTROL switches set to PULSE, attitude hold mode is not affected. With GUID CONT switch set to PGNS and one or more ATTITUDE CONTROL switches set to DIR, ACA displacement of one-fourth of full throw causes continuous two-jet firing for selected axes, without rate feedback interfering with LGC command signals.

Location/ Ref Des	Name and Position	Function	Circuit Breaker	Power Source	Remarks
Panel 3 (cont)	ALARM/TONE	Tests lamps in MASTER ALARM pushbuttons, and tests audible warning tone.	LTG: MASTER ALARM (panel 16)	LMP's 28-vdc bus	
	C/W 1	Tests lamps in first bank of warning lights on panel 1.			
	ENG PB-C/W 2	Tests lamps in engine START and STOP lights and in second bank of warning lights on panel 1.			
	C/W 3	Tests lamps in first bank of caution lights on panel 2.			
	C/W 4	Tests lamps in second bank of caution lights on panel 2.			
	COMPNT	Tests lamps in component caution and lunar contact lights.			
16S17	EXTERIOR LTG switch	Controls docking and tracking lights.	LTG: ANN/ DOCK/ COMPNT (panel 11 and panel 16)	CDR's 28-vdc bus	
	DOCK	Energizes docking lights.			
	OFF	Deenergizes docking and tracking lights.			
	TRACK	Energizes tracking light to full (100%) brightness.			
1DS2	LUNAR CONTACT light	Both lights (red) go on when probe(s) on landing gear touches lunar surface, to indicate that descent engine should be turned off.	N/A	N/A	Light goes off when either engine stop switch is pressed.

Location/ Ref Des	Name and Position	Function	Circuit Breaker	Power Source	Remarks
Panel 3 (cont) 9S18	X-POINTER SCALE switch	Controls scale range of X-pointer indicator.	N/A	N/A	
	HI MULT LO MULT	When LOS azimuth and elevation rates are displayed, this position provides a scale of ± 20 mrad/sec. When horizontal velocities are displayed, this position illuminates the X10 multiplier to provide range of ± 200 fps. When LOS azimuth and elevation rates are displayed, this position illuminates the X.1 multiplier to provide a range of ± 2 mrad/sec. When horizontal velocities are displayed, this position provides range of ± 20 fps.			

Mission LM Basic Date 1 February 1970 Change Date _____ PANEL 3
 Page 3-73/3-74

Mission LM Basic Date 1 February 1970 Change Date _____
PANEL 4
Page 3-75

Location/ Ref Des	Name and Position	Function	Circuit Breaker	Power Source	Remarks
Panel 4 1S21 1S23 1S20 1S22 3DS1 3DS8	ACA/4 JET switch (CDR and LMP) ENABLE DISABLE	Enables isolation of 28 volts dc if short occurs in ACA. Enables normal operation of ACA. Disables four-jet direct attitude control function (hardover position) of ACA.	STAB/CONT; ATT DIR CONT (panel 11)	CDR's 28-vdc bus	The nominal position of this switch is ENABLE.
	TTCA/TRANSL switch (CDR and LMP) ENABLE DISABLE	Enables isolation of 28 volts dc if short occurs in TTCA. Enables normal operation of TTCA. Disables translation control function of TTCA.	STAB/CONT; ATCA (AGS) (panel 16)	LMP's 28-vdc bus	The nominal position of this switch is ENABLE.
	NOTE White condition indicators signify computer status; yellow condition indicators describe various cautions associated with the GN&CS.				Is energized by first character of digital uplink message received by LGC. If light is not turned off by end of uplink transmission, the astronaut should extinguish light when transmission is complete by pressing RSET pushbutton.
	3DS1 UPLINK ACTY condition indicator (white)	Indicates that uplink is being received by LGC.	N/A	N/A	
	3DS8 TEMP condition indicator (yellow)	Indicates that stable member temperature has exceeded its limits by $\pm 4^{\circ}$ F. (Nominal operating temperature $+130.3^{\circ}$ F).	N/A	N/A	

PANEL
4

Location/ Ref Des	Name and Position	Function	Circuit Breaker	Power Source	Remarks	
Panel 4 (cont)	3DS5	NO ATT condition indicator (white)	Indicates that ISS is not suitable for use as attitude reference.	N/A	N/A	Indicator lamp goes on when LGC is in operate mode and there is no inertial reference, during coarse align, if ISS is off, or IMU is caged.
	3DS9	GIMBAL LOCK condition indicator (yellow)	Indicates that middle gimbal has driven through angle greater than +70° or -70° from its zero position.	N/A	N/A	When middle gimbal angle exceeds +85° or -85° from its zero position LGC automatically commands coarse align mode in ISS to prevent gimbal oscillation. NO ATT condition indicator goes on.
	3DS6	STBY condition indicator (white)	Indicates that LGC is in standby condition.	N/A	N/A	Indicator lamp is deenergized if LGC is in operate mode.
	3DS10	PROG condition indicator (yellow)	Indicates that program check has failed.	N/A	N/A	Indicator is controlled by computer program.
	3DS7	KEY REL condition indicator (white)	Indicates that internal program has attempted to use DSKY and found it busy due to independent operator action.	N/A	N/A	Indicator is modulated by flash signal.
	3DS11	RESTART condition indicator (yellow)	Indicates one or more of the following abnormal operations: Word has been transferred incorrectly from memory-parity fail. LGC is in endless control loop-TC trap. LGC has been interrupted for 30 msec - RUPT lock. Alarm has been generated by program control. LGC fails to process new job within a period from .64 to 1.92 seconds.	N/A	N/A	

Location/ Ref Des	Name and Position	Function	Circuit Breaker	Power Source	Remarks
Panel 4 (cont)					
3DS14	OPR ERR condition indicator (white)	Indicates that astronaut has performed an improper sequence on the DSKY.	N/A	N/A	Goes out when RSET pushbutton is pressed.
3DS12	TRACKER condition indicator (yellow)	Indicates RR CDU failure or improper data from RR or LR.	N/A	N/A	When RR or LR are on, indicator lamp is energized if: RR CDU fine error greater than 1.0 vrms occurs RR CDU coarse error greater than 2.5 vrms occurs RR read counter limit cycles at rate greater than 160 cps RR CDU analog-to-digital conversion tails RR CDU +14-vdc supply decreases to 50% of nominal RR data-not-good discrete occurs during LGC data read sequence LR altitude data-not-good discrete or LR velocity data-not-good discrete occurs during LGC data read sequence.
3DS209	ALT light (yellow)	Indicates altitude data-good signal from LR is absent during Landing Radar Data Read Routine.	N/A	N/A	A flashing light indicates that LGC received data-good signal, but data does not pass LGC Reasonableness Test.
3DS204	VEL light (yellow)	Indicates velocity data-good signal from LR is absent during Landing Radar Data Read Routine.	N/A	N/A	

Location/ Ref Des	Name and Position	Function	Circuit Breaker	Power Source	Remarks
Panel 4 (cont)					
3DS15	COMPTR ACTY condition indicator (white)	Indicates that LGC is in program other than dummy job (LGC is processing data.)	N/A	N/A	
3M1	PROG display indicator	Indicates the program that is being processed LGC.	N/A	N/A	
3M2	VERB display indicator	Indicates verb code entered at key- board or commanded by LGC.	N/A	N/A	
3M3	NOUN display indicator	Indicates noun code entered at key- board or commanded by LGC.	N/A	N/A	
3M4 3M5 3M6	Data display registers	Indicate numerical data entered at keyboard or commanded by LGC. Sign is associated only with decimal data.	N/A	N/A	The three data display registers are designated R1, R2, R3 from top to bottom of DSKY.
3S3	VERB pushbutton (momentary)	Conditions LGC to interpret next two numerical characters as verb code and extinguishes verb numer- ical display.	N/A	N/A	
3S4	NOUN pushbutton (momentary)	Conditions LGC to interpret next two numerical characters as noun code and extinguishes noun numer- ical display.	N/A	N/A	
3S5	CLR pushbutton (momentary)	Used during data loading to blank data register being used; loading can then be resumed.	N/A	N/A	First component load verbs are cleared by pressing CLR pushbut- ton once. Second and third compo- nent load verbs are cleared by pressing the pushbutton twice or three times, as required. The data registers clear in R3, R2, R1 order. Refer to paragraph 2.2.3.1.3.

Location/ Ref Des	Name and Position	Function	Circuit Breaker	Power Source	Remarks
Panel 4 (cont)					
3S6	PRO pushbutton (momentary)	Commands LGC to proceed without data. Also commands LGC to enter standby mode if pressed when program P06 is in progress.	N/A	N/A	
3S7	KEY REL pushbutton (momentary)	Releases DSKY to LGC for use with internal sequences.	N/A	N/A	Operator should release DSKY by pressing pushbutton when he is temporarily finished with DSKY.
3S8	ENTR pushbutton (momentary)	Informs LGC that assembled data are complete and that requested function is to be executed. Used in two ways: to execute the VERB/ NOUN code appearing on the VERB and NOUN display and to accept data just loaded and appearing in data display registers (R1, R2, R3).	N/A	N/A	
3S9	RSET pushbutton (momentary)	Turns off DSKY condition indicator after condition has been corrected.	N/A	N/A	
3S10 thru 3S19	0 through 9 pushbuttons (momentary)	Enter numerical data, noun codes, and verb codes into DSKY displays.	N/A	N/A	
3S21 3S20	+ and - pushbutton (momentary)	Inform LGC that following numerical data are decimal and indicate sign of data.	N/A	N/A	
3DS	PR10 DISP light	Indicates that system has a priority display to bring to the attention of the astronaut.	N/A	N/A	
3DS	NO DAP light	Indicates that the DAP is not controlling vehicle attitude.	N/A	N/A	

Mission LM Basic Date 1 February 1970 Change Date
PANEL 5
Page 3-81

Location/ Ref Des	Name and Position	Function	Circuit Breaker	Power Source	Remarks
Panel 5					
1S5	DES RATE switch + 1 FPS center - 1 FPS	Permits establishing LM rate of descent under PGNS control in fixed increments. Each switch actuation provides a discrete pulse, changing the rate of descent by 1 fps.	N/A	N/A	Response to switch commands is monitored on the altitude rate scale of the Range/Range Rate; Alt/Alt Rate indicator. Plus sign denotes increase in engine thrust. Minus sign denotes decrease in engine thrust.
1S3	Engine stop pushbutton	Provides discrete stop signals to descent and ascent engines, independent of setting of ENG ARM switch, except when in abort sequence.	STAB/CONT: ENG ARM (panel 16)	LMP's 28-vdc bus	When pressed, pushbutton remains pressed and illuminated (red). It is reset by pressing it again.
1S2	Engine START pushbutton	Permits immediate manual firing of descent or ascent engine, depending upon setting of ENG ARM switch.	STAB/CONT: ENG START OVRD (panel 11)	CDR's 28-vdc bus	When pressed, pushbutton remains pressed and illuminated (red). Actuation of either engine stop pushbutton interrupts override function, the engine stops, and the START pushbutton light goes out.
1S12	+X TRANSL pushbutton (momentary)	When held pressed, switch provides four-jet translation in +X-direction by energizing RCS direct (secondary) coils. Upon release of pushbutton, signal is removed from coils, terminating jet firings.	STAB/CONT: ATT DIR CONT (panel 11)	CDR's 28-vdc bus	If switch fails closed, circuit breaker must be opened. The secondary coils of the RCS engines will be disabled, resulting in loss of hardover and direct modes.
9S19	MISSION TIMER: TIMER CONT switch	Provides discrete signals to MISSION TIMER indicator.	FLIGHT DISPLAYS: MISSION TIMER (panel 11)	CDR's 28-vdc bus	
AA	START	Enables indicator to count upward from number preset with SLEW CONT switches.			

PANEL
5

Location/ Ref Des	Name and Position	Function	Circuit Breaker	Power Source	Remarks
Panel 5 (cont)	STOP	Stops and holds indicator count until START or RESET is selected.			
	RESET	Resets indicator to zero.			
	9S22	SLEW CONT HOURS switch	FLIGHT DISPLAYS: MISSION TIMER (panel 11)	CDR's 28-vdc bus	Hundreds-column digit of HOURS columns is also controlled by the tens position, at rate of one digit per 5 seconds. All digits change in increasing direction.
	TENS	Slews tens-column digit of HOURS columns at rate of two digits per second as long as switch is held in this position.			
	Center	Disables slewing function.			
	UNITS	Slews units-column digit of HOURS columns at rate of two digits per second as long as switch is held in this position.			
	9S21	SLEW CONT MIN switch	FLIGHT DISPLAYS: MISSION TIMER (panel 11)	CDR's 28-vdc bus	All digits change in increasing direction.
	TENS	Slews tens-column digit of MIN columns at rate of two digits per second as long as switch is held in this position.			
	Center	Disables slewing function.			
	UNITS	Slews units-column digit of MIN columns at rate of two digits per second as long as switch is held in this position.			

Mission LM Basic Date 1 February 1970 Change Date PANEL 5 Page 3-83

Location/ Ref Des	Name and Position	Function	Circuit Breaker	Power Source	Remarks
Panel 5 (cont) 9S20	SLEW CONT SEC switch	Selects inputs to MISSION TIMER indicator.	FLIGHT DISPLAYS: MISSION TIMER (panel 11)	CDR's 28-vdc bus	
	TENS	Slews tens-column digit of SEC columns at rate of two digits per second as long as switch is held in this position.			
	Center	Disables slewing function.			
	UNITS	Slew units-column digit of SEC columns at rate of two digits per second as long as switch is held in this position.			
16S4	OVERRIDE ANUN switch	Provides full voltage bypass of C/W array and component caution light portion of ANUN/NUM control.	N/A	N/A	
	ON	Bypasses ANUN/NUM control and applies 5 volts dc directly to annunciators of C/W array, to component caution lights, and to incandescent illuminated pushbuttons.			
16S5	OFF	C/W array and component caution lights are controlled with ANUN/NUM control.	AC BUS B: NUM LTG: (panel 11)	CDR's 115-vac bus	Portion of autotransformer associated with numeric readouts is disconnected from circuit.
	OVERRIDE NUM switch	Provides full voltage override of brightness control of numeric portion of ANUN/NUM control.			
	ON	Supplies full power (115 volts ac) directly to integrally illuminated EL numerics and time-shared displays.			
	OFF	Disconnects full power; EL numeric readouts are controlled with ANUN/NUM control.			

PANEL
5

Page 3-84 Mission LM Basic Date 1 February 1970 Change Date

Location/ Ref Des	Name and Position	Function	Circuit Breaker	Power Source	Remarks
Panel 5 (cont)	16S3	OVERRIDE INTEGRAL switch ON OFF	AC BUS A: INTGL LTG: (panel 11) AC BUS A: INTGL LTG: (panel 11) LTG:FLOOD (panel 16)	CDR's 115-vac bus CDR's 115-vac bus LMP's 28-vdc bus	Full voltage is 115 volts ac, 400 cps.
	16S1	SIDE PANELS switch ON OFF			
	16R1	FLOOD OVHD/FWD control DIM to BRIGHT			

Location/ Ref Des	Name and Position	Function	Circuit Breaker	Power Source	Remarks
Panel 5 (cont)	16R2	ANUN/NUM control	LTG:NUM (panel 11)	CDR's 115-vac bus	If either portion of ANUN/NUM control fails, separate override switch provides maximum voltage for annunciators and component caution lights (ANUN OVERRIDE switch) and for EL numeric readouts (NUM OVERRIDE switch). Maximum voltage supplied on BRIGHT position: 5 volts dc for annunciators and component caution lights; 115 volts ac for numeric readouts.
	16R3	INTEGRAL control	LTG: INTGL (panel 11)	CDR's 115-vac bus	If control fails or brightness range is inadequate, control may be bypassed with INTEGRAL OVERRIDE switch.

Mission LM Basic Date 1 February 1970 Change Date PANEL 6 Page 3-87

Location/ Ref Des	Name and Position	Function	Circuit Breaker	Power Source	Remarks
Panel 6					
184	Engine stop pushbutton	Provides discrete stop signals to descent and ascent engines, independent of setting of ENG ARM switch, except when in abort sequence.	STAB/CONT: ENG ARM (panel 16)	LMP's 28-vdc bus	Pushbutton is pressed on lunar landing, after LUNAR CONTACT lights go on. When pressed, pushbutton remains pressed and illuminated red. It is reset by pressing it again.
12S17	AGS STATUS switch OFF STANDBY OPERATE	Enables astronauts to place AGS in any one of three states. No power is supplied to AGS. Supplies power to ASA heaters. Applies full power to entire ASA and then places AEA in warmup mode. Applies full power to ASA and AEA.	STAB/CONT: AEA (panel 16) STAB/CONT: ASA (panel 16)	LMP's 28-vdc bus	
12DS1	OPR ERR condition indicator (white)	Goes on if astronaut presses incorrect number of pushbuttons for operation being performed, if + or - keys are not pressed first in entering data, and if "8" or "9" is used in entering address (which is limited to octal digits).	N/A	N/A	If too few pushbuttons are pressed, indicator goes on after ENTR or READOUT pushbutton is pressed. If too many pushbuttons are pressed, indicator goes on before ENTR or READOUT pushbutton is pressed. Indicator goes off when CLR pushbutton is pressed.
12M1	Address indicator	Displays digits that identify AEA computer address selected by astronaut via keyboard.	N/A	N/A	Three octal digits must be set on this indicator, using keyboard, before each DEDA operation.
12M2	Data indicator	Displays data to be entered into AEA computer address displayed on address indicator. Displays data selected for readout from AEA computer location identified on address indicator, if READOUT pushbutton is pressed.	N/A	N/A	Five-digit numeric indicator and + and - indicator. Data to be entered are selected by astronaut via keyboard. Updating rate of any changing data selected for readout is limited to two times per second.

PANEL 6

Location/ Ref Des	Name and Position	Function	Circuit Breaker	Power Source	Remarks
Panel 6 (cont)					
12S1 thru 12S12	Keyboard pushbuttons	Enable astronauts to load shift register with necessary values (0 through 9) and signs (+ and -) to be transferred to AEA.	STAB/CONT: AEA (Panel 16)	LMP's 28-vdc bus	Pressing pushbuttons successively fills address and data indicators in order, from left to right.
		NOTE If the CLR, ENTR, or READOUT pushbuttons are not fully depressed, either the OPR ERR light will illuminate or the DEDA will not operate.			
12S13	CLR pushbutton (momentary)	Erases all numerals displayed on address and data indicators and cancels effect of HOLD switch.	N/A	N/A	When pressed, switch extinguishes OPR ERR indicator lamp.
12S15	READOUT pushbutton (momentary)	Enables astronauts to call up AEA computer data from address set up on address indicator, for display on data indicator.	N/A	N/A	
12S14	ENTR pushbutton (momentary)	Enables astronauts to enter data (set up on data indicator) into AEA computer at address set up on address indicator.	N/A	N/A	Address and data indicators are cleared for immediate reuse when AEA computer has accepted data.
12S16	HOLD pushbutton (momentary)	Enables astronauts to interrupt updating of any continuously changing data displayed on data indicator.	N/A	N/A	Information displayed on data indicator remains unchanged until pushbutton is pressed again, enabling display of present value of data and continuance of updating.

Page 3-152 Mission LM Basic Date 1 February 1970 Change Date MISC

Location/ Ref Des	Name and Position	Function	Circuit Breaker	Power Source	Remarks
ORDEAL (cont)	INRTL	Enables CDR's FDAI to display total attitude with respect to IMU.			
9S24	2 ORB RATE INRTL	Selects input to LMP's FDAI. Enables LMP's FDAI to display total attitude with respect to local horizontal reference frame. Enables LMP's FDAI to display total attitude with respect to IMU.	N/A	N/A	
9S25	EARTH-LUNAR switch EARTH PWR OFF LUNAR	Used in conjunction with ALT SET control, inserts appropriate scale factor into ORDEAL electronics. Supplies proper scale factor to ORDEAL electronics when LM is in earth orbit. Removes ac and dc power from ORDEAL. Same as EARTH position, except that scale factor applies to lunar orbit.	AC BUS B; ORDEAL (panel 11) FLIGHT DISPLAYS; ORDEAL (panel 11)	CDR's 115-vac bus B CDR's 28-vdc bus	Switch is always set to either LUNAR or OFF.
9R1	ALT SET control	Inserts LM orbital altitude into ORDEAL.	N/A	N/A	Control can be set from 10 to 310 nautical miles.
9S26	LIGHTING switch BRT OFF DIM	Controls the brightness of the ORDEAL EL panel lighting. Selects bright light level. Turns off panel lighting. Selects dim light level.	AC BUS B; ORDEAL (panel 11)	CDR's 115-vac bus B	

Location/ Ref Des	Name and Position	Function	Circuit Breaker	Power Source	Remarks
ORDEAL (cont) 9S27	MODE switch	Inserts and maintains proper angular offset into FDAI's.	N/A	N/A	
	OPR/SLOW	When used with UP and DOWN positions of the SLEW switch, permits slewing the FDAI at 10X orbital rate in plus or minus direction. Used for fine adjustment.			
	HOLD/FAST	When used with UP and DOWN positions of SLEW switch, permits slewing FDAI at 200X orbital rate in plus or minus direction. Used for coarse adjustment.			
9S28	SLEW switch	Slews pitch indication on FDAI in plus (UP position) or minus (DOWN position) at rate dependent on setting of MODE switch.			
UTILITY LIGHT 16S19 16S20	UTILITY LIGHTS switch (CDR and LMP)	Controls outlet that accepts portable utility light for use in cabin.	LTG UTIL (Panel 11)	CDR's 28-vdc bus	
	OFF	Deenergizes utility light.			
	DIM	Provides reduced illumination.			
	BRIGHT	Provides maximum illumination.			
17S9	RR GYRO SEL switch		N/A		
	PRIM	Inhibits voting logic and selects primary gyros.			
	AUTO	Enables voting logic to select primary or secondary gyros automatically.			
	SEC	Inhibits voting logic and selects secondary gyros.			

LMA790-3-LM
APOLLO OPERATIONS HANDBOOK
CONTROLS AND DISPLAYS

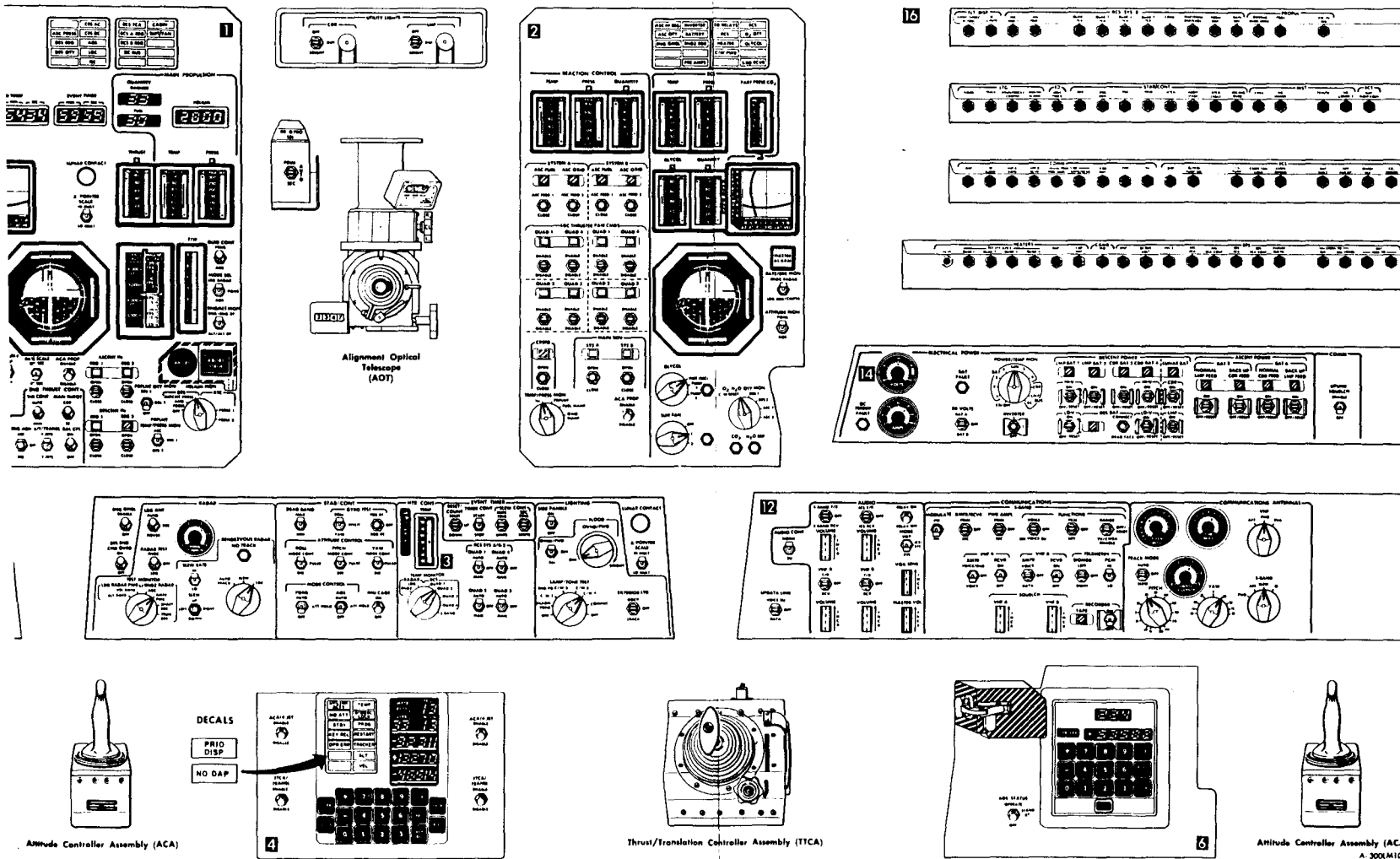


Figure 3-2. Controls and Displays (LM 8 and LM 9)